

ECS7013P: Deep Learning for Audio and Music

Lab 2 – Receptive Field



Figure 1 A least bittern

Part 1

For the *least bittern* “coo coo coo” breeding call:

- 0.5 seconds long, 500-600 Hz
- Recorded at 44.1 kHz
- Spectrogram: frame size of 1024 samples, hop size of 512 samples

(1) Calculate: bandwidth of a single spectrogram bin; spectrogram frame rate; dimensions of the “rectangle” containing the call.

(2) For a simple network:

$3 \times 3 \text{ conv} \rightarrow 2 \times 2 \text{ maxpool} \rightarrow 3 \times 3 \text{ conv}$

- Calculate the receptive field of a pixel in the output layer.
- Does it accommodate the least bittern call?



Figure 2 A red-whiskered bulbul

Part 2

Do it similar to Part 1 using the settings as used in the “bulbul” bird detection network ([Grill & Schlueter 2016](#)):

- Recordings downsampled to 22.05 kHz
- Spectrogram: frame size of 1024 samples, hop size of 315 samples

Network architecture:

3×3 conv → 3×3 maxpool →
3×3 conv → 3×3 maxpool →
3×1 conv → 3×1 maxpool →
3×1 conv → 3×1 maxpool